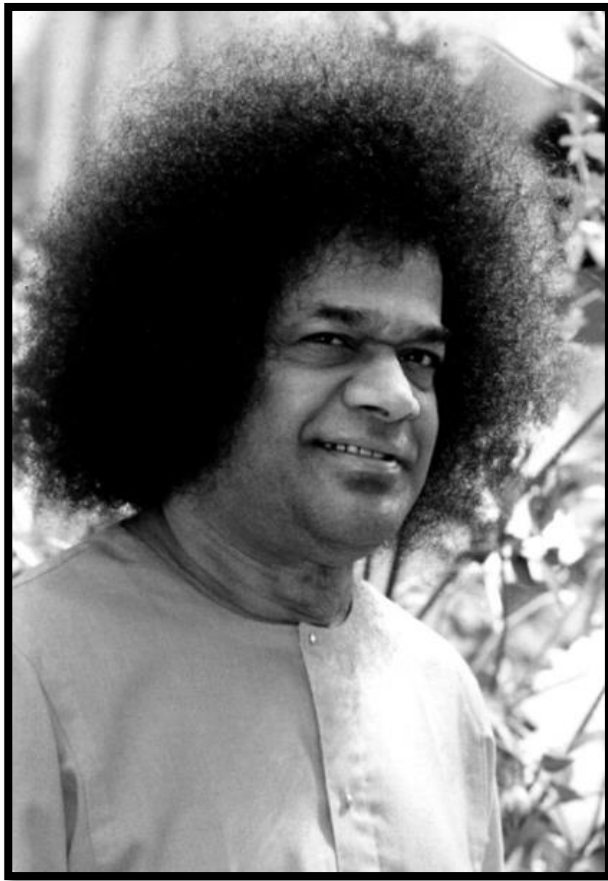




Multi-modal Cattle Verification Using Vision Transformers and Cross- Attention Refinement

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Problem Statement and Introduction

overview & proposed solution

Why Cattle Verification?

- Disease traceability & livestock management
- Insurance validation & ownership authentication
- Food supply chain transparency
- Theft prevention
- 1:1 matching — does this sample belong to this animal?
- Different from identification (1:N closed-set)

Limitations of Existing Methods

- Unreliable Traditional Methods
- Single-modality: vulnerable to occlusion & pose variation
- Naive fusion (concat/averaging) — misses cross-modal cues
- Triplet loss: slow convergence, suboptimal representations

Proposed Two-Stage Framework

Stage 1	Shared ViT-Base/16 backbone learns modality-invariant 512-dim embeddings from face
Stage 2	Lightweight Cross-Attention Refinement (CAR) module models bidirectional face $\leftrightarrow$ muzzle dependencies.
Loss	Multi-Similarity Loss with Hard Pair Mining

Datasets Used



Cattle-1

Shojaeipour et al., 2021

Animals: 300

Face: 2,898

Muzzle: 2,804

Images: 5,702

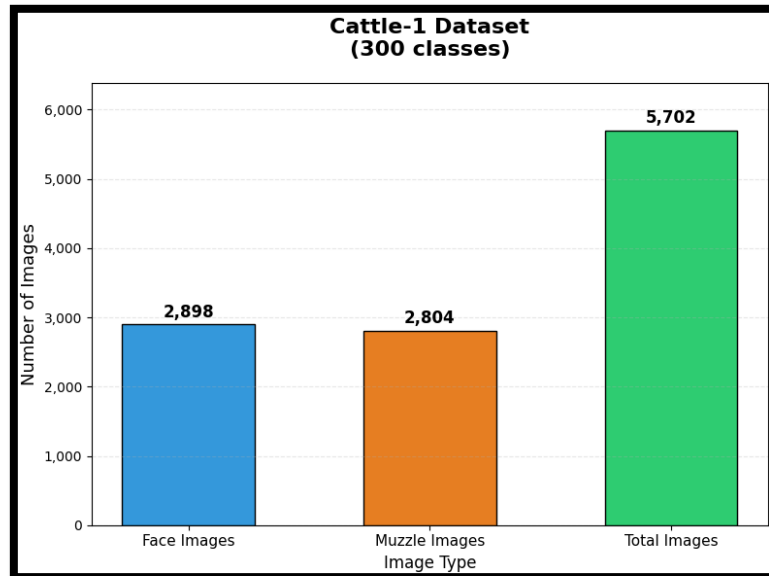


Fig. 1: Cattle-1 – Shojaeipour et al., 2021, agronomy

Cattle-2

Ahmed et al., 2024

Animals: 459 (456 used)

Face: 2,763

Muzzle: 2,545

Images: 5,308

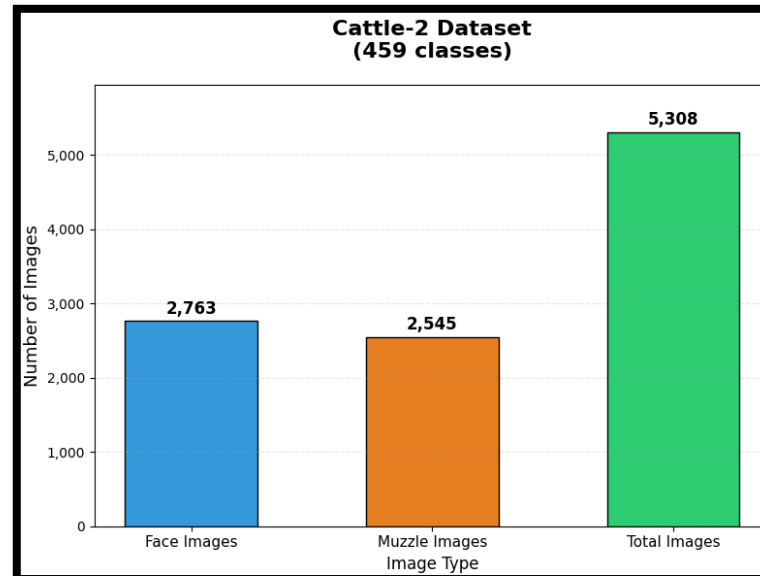


Fig. 2: Cattle-2 – Ahmed et al., 2024, Data in Brief

Cattle-3 (In-house)

Our dataset

Animals: 88

Face: 597

Muzzle: 950

Images: 1,547

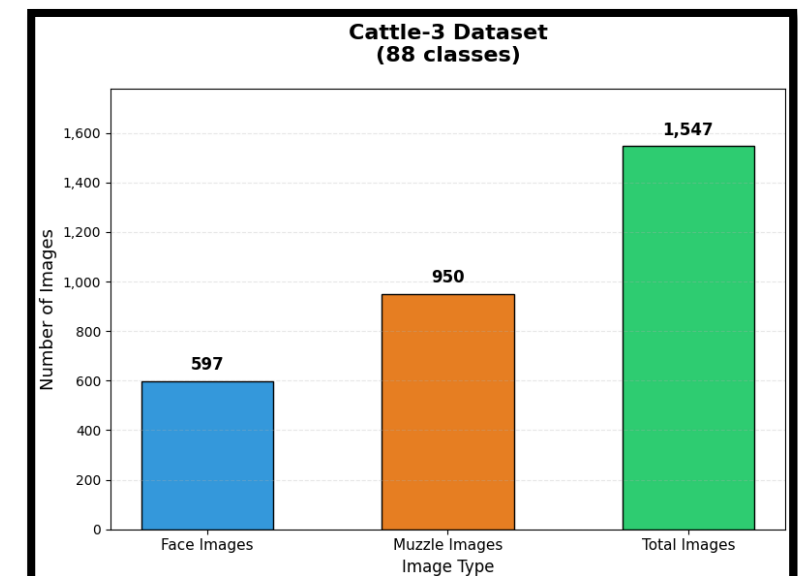
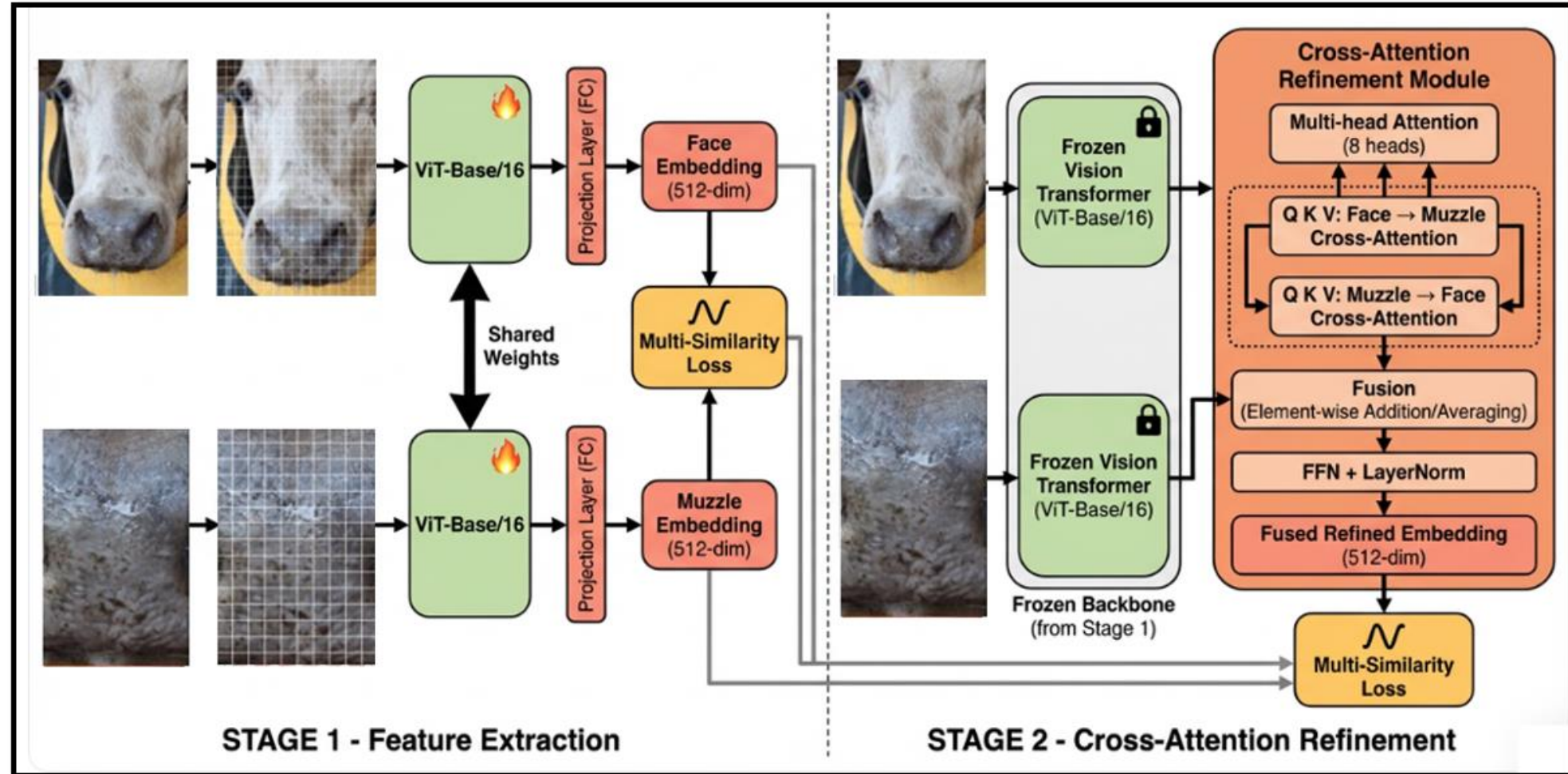


Fig. 3: Cattle-3 – In-house dataset

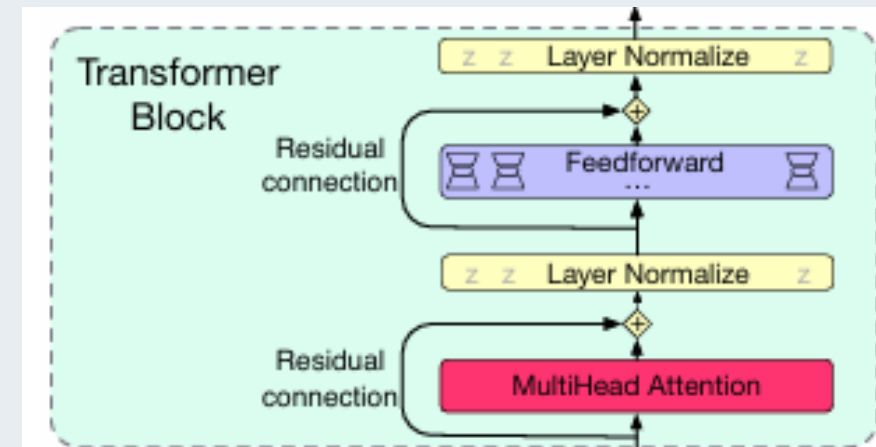
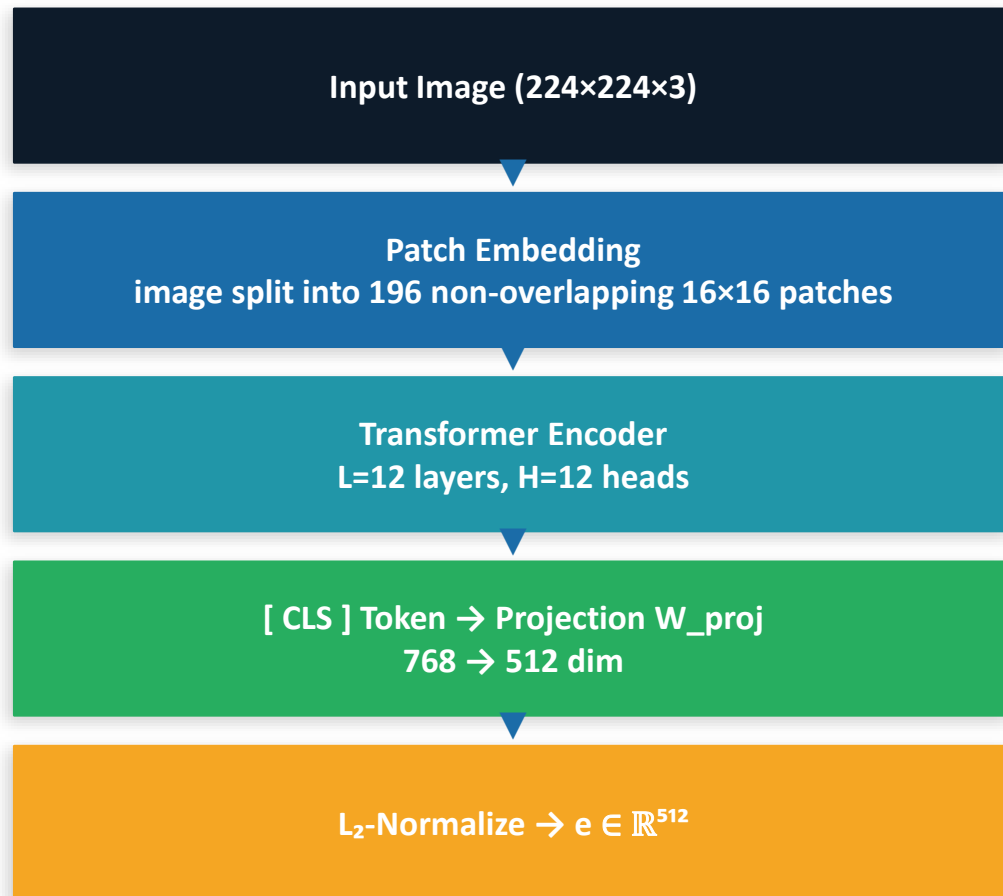
METHODOLOGY:



Stage 1 — Global Feature Extraction via ViT



Vision Transformer Architecture



Final embedding:

$$e = L_2\text{-norm}(W_{\text{proj}} \cdot z_0^{(L)}) \in \mathbb{R}^{512}$$

Source: Jurafsky & Martin, *Speech and Language Processing*, 3rd ed.

Multi-Similarity Loss with Hard Pair Mining



✗ Triplet Loss

Equal weight to all triplets → slow convergence, gradient vanishing, suboptimal at scale

Why Multi-Similarity Loss?

- Triplet loss only considers ONE positive and ONE negative per anchor
- MS Loss uses ALL hard positives & hard negatives simultaneously
- Adaptive pair-wise weights → richer gradient signals
- Logarithmic formulation prevents gradient vanishing
- Faster convergence and better embedding quality

✓ Multi-Similarity Loss

Adaptive pair weights based on 3 similarity relations → non-zero gradient for ALL hard pairs

Hard Pair Mining

Hard Positives — positives where a negative is closer:

$$P_{\text{hard}}^i = \left\{ j \in P_i \mid S_{ij} > \min_{k \in N_i} S_{ik} - \varepsilon \right\}$$

Hard Negatives — negatives closer than some positive:

$$N_{\text{hard}}^i = \left\{ k \in N_i \mid S_{ik} < \max_{j \in P_i} S_{ij} + \varepsilon \right\}$$

MS Loss Formulation

$$\mathcal{L}_i = \frac{1}{\alpha} \log \left(1 + \sum_{j \in P_{\text{hard}}^i} \exp(-\alpha(S_{ij} - \lambda)) \right) + \frac{1}{\beta} \log \left(1 + \sum_{k \in N_{\text{hard}}^i} \exp(\beta(S_{ik} - \lambda)) \right)$$

Hyperparameters used: $\alpha = 2.0$, $\beta = 40.0$, $\lambda = 0.5$

In Stage 2: loss computed jointly over face, muzzle & fused embeddings

Stage 2: Cross-Attention Refinement (CAR)

Why Cross-Attention?

Stage 1 creates a shared embedding space but does NOT model the interaction between face and muzzle of the same animal.

CAR module explicitly captures bidirectional dependencies — what does the face tell us about the muzzle, and vice versa?

Inputs: $f \in \mathbb{R}^{512}$, $m \in \mathbb{R}^{512}$ from frozen backbone

$$f \rightarrow m = \text{MHA}(Q=f, K=m, V=m, H=8)$$

$$m \rightarrow f = \text{MHA}(Q=m, K=f, V=f, H=8)$$

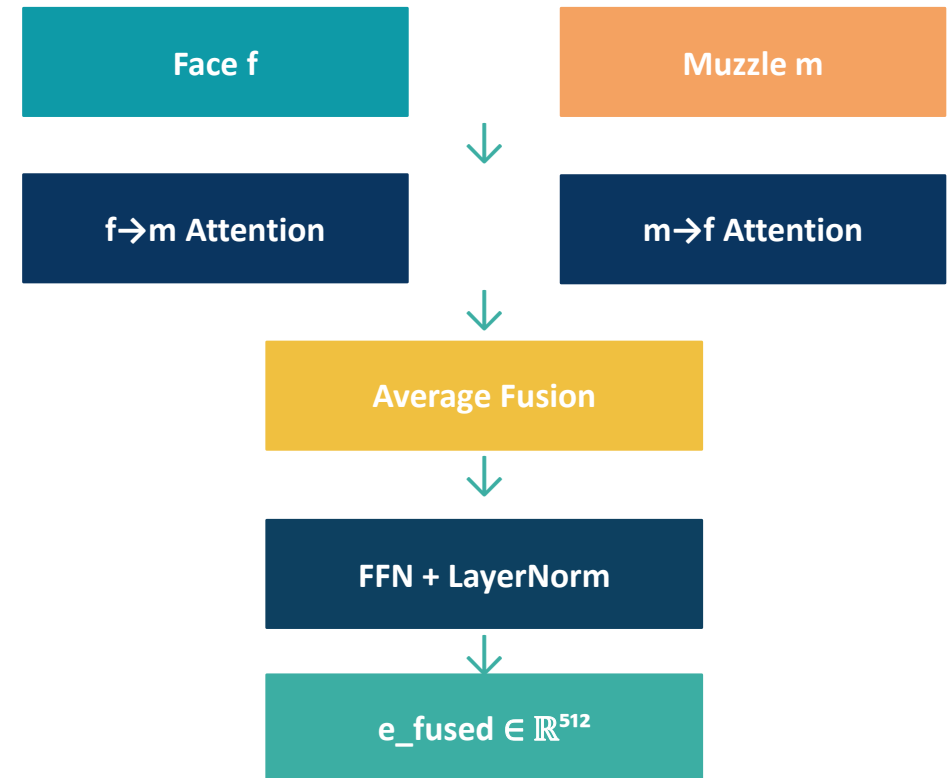
$$f_fused = (f \rightarrow m + m \rightarrow f) / 2$$

$$s(F, M) = \theta(F) \cdot \theta(M), \quad \text{Decision}(F, M) = \begin{cases} \text{Accept} & s \geq \tau^* \\ \text{Reject} & s < \tau^* \end{cases}$$

The threshold τ^* is the Equal Error Rate (EER) point on the validation set, where $\text{FAR}(\tau) = \text{FRR}(\tau)$.

Backbone FROZEN in Stage 2
Preserves globally discriminative Stage-1 features while CAR learns cross-modal refinement

Information Flow in CAR



Verification Results



Metric	Stage-1 Cattle-1	Stage-2 Cattle-1	Stage-1 Cattle-2	Stage-2 Cattle-2	Stage-1 Cattle-3	Stage-2 Cattle-3
Accuracy	95.16	97.13	97.71	98.67	92.46	95.21
Precision	95.17	97.11	97.71	98.69	92.46	95.38
Recall	95.14	97.11	97.71	98.66	92.46	95.02
F1 Score	95.16	97.14	97.71	98.67	92.46	95.20
AUC	98.96	99.70	99.78	99.87	96.54	98.93

Comparison with biometric methods.



Method	Feature	Accuracy (%)
Awad & Hassaballah (2019)	Muzzle	75.70
Li et al. (2022)	Muzzle	91.00
Gaber et al. (2022)	Muzzle	71.80
Lee et al. (2023)	Muzzle	92.10
Xu et al. (2022)	Face	91.30
Bergman et al. (2024)	Face	96.30
Kumar et al. (2025)	Face+Muzzle	93.67
Ours — Stage 1	Face+Muzzle	95.16
Ours — Stage 2 ✓	Face+Muzzle	97.13

Comparison with State of the Art methods.

Method	Cattle-1	Cattle-2	Cattle-3
Cattle Verification with Pair wise triplet loss Kumar et al. (2025)	94.42	93.21	89.37
Multi-modal Cattle verification (ours)	97.13	98.67	95.21

Ablation Study — Unimodal vs Multi-modal

Face-Only vs. Muzzle-Only vs. Multi-modal (Stage 2) — Accuracy %

Dataset	Cattle-1 Face	Cattle-1 Muzzle	Cattle-1 Multi ✓	Cattle-2 Face	Cattle-2 Muzzle	Cattle-2 Multi ✓	Cattle-3 Face	Cattle-3 Muzzle	Cattle-3 Multi ✓
Accuracy	97.72	97.00	97.13	99.05	97.31	98.67	94.22	91.63	95.21
AUC	99.81	99.66	99.70	99.96	99.33	99.87	98.83	95.28	98.93

Key Insights

Face > Muzzle in isolation

Face-only consistently outperforms muzzle-only. Muzzle quality can be affected by imperfect YOLO11 extraction and background noise.

Multi-modal wins on hard data

On challenging Cattle-3 (Indian breeds, varied lighting), multi-modal fusion improves over face-only: 95.21% vs 94.22%. CAR leverages complementarity when both modalities are less reliable.

Fusion benefit depends on quality

On Cattle-1 and Cattle-2, face-only slightly edges out fusion — but Stage-2 multi-modal still achieves the highest accuracy by combining all signals.



Cattle-1 (A),

Cattle-2 (B),

Cattle-3 (C)

- Consistent attention patterns across all three datasets confirm that the ViT backbone has learned **modality-appropriate discriminative features** rather than dataset-specific artefacts.
- This qualitative validation corroborates the quantitative results — the model is attending to the same regions a domain expert would identify as most informative for biometric identification.

Conclusion and Future Work



Contributions

- 1 Two-stage multi-modal framework: shared ViT backbone + CAR module
- 2 Multi-Similarity Loss with hard pair mining — outperforms triplet loss
- 3 Automated YOLO11 muzzle extraction — no manual preprocessing
- 4 State-of-the-art on 3 datasets: up to 98.67% acc. & 99.87% AUC
- 5 In-house Cattle-3: first verification benchmark for Indian cattle breeds

Limitations & Future Work

- ⚠ **Small identity pool**
→ Evaluate on larger, more diverse cattle populations
- ⚠ **Requires both modalities**
→ Develop single-modal fallback when muzzle unavailable
- ⚠ **Low-light not explored**
→ Test under night-time, motion-blur conditions
- ⚠ **Compute-intensive ViT**
→ Integrate MobileViT for edge/farm device deployment
- ⚠ **Separate preprocessing**
→ Fully integrated end-to-end pipeline with YOLO embedded



Paper Submission

Paper Title :

Multi-modal Cattle Verification Using Vision Transformers and Cross-Attention Refinement

Conference Name :

National Conference on Computer Vision, Pattern Recognition, Image Processing and Graphics

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THANK YOU!

